
Conditioning Physics-Informed Surrogates on Initial Conditions: A Query-Aligned Functional Encoder for Parameterized PDEs

Anonymous Authors
Paper 19 Research Group
anonymous@example.org

Abstract

Physics-informed neural networks (PINNs) that amortize across a *family* of PDEs have so far been conditioned on the PDE coefficients alone: the initial and boundary conditions enter only through the loss and are frozen per trained model, so a single surrogate cannot be *queried* at a new initial condition. We extend the parameter encoder of a coefficient-conditioned PINN into a *problem-specification encoder* that ingests the initial condition as a functional input—a 16-point discretization whose truncated Fourier coefficients form a global specification code, and whose trigonometric interpolant is evaluated *at the query coordinate* inside the forward pass. On a reaction family (logistic growth, non-linear in the initial condition) with 130 trained coefficient–initial-condition specifications, a single model is evaluated at more than 300 additional specifications spanning unseen initial-condition parameters, unseen parameter combinations, unseen functional families, unseen coefficients, and their cross-products. The query-aligned functional encoder beats a capacity- and information-matched control that receives the same samples as raw extra inputs on unseen parameters (0.092 vs. 0.167), unseen within-family combinations (0.039 vs. 0.272), unseen coefficient×IC compositions (0.175 vs. 0.250), and out-of-distribution probes (0.339 vs. 0.501), while the control wins on an unseen compactly supported family and on coefficient interpolation—a split we explain through the geometry of the specification code. Conditioning is validated by shuffled-, zeroed-, and constant-code ablations, a latent-space analysis that connects where the encoder generalizes to where the code interpolates, and a 3×-budget probe that separates optimization limits from representation limits. A single multi-family model jointly supports convection, diffusion, and reaction specifications and transfers to unseen compositions, while transfer to convection–diffusion–reaction equations absent from training is an honest negative.

1 Introduction

Parametric PDE surrogates are motivated by multi-query workflows—design optimization, uncertainty quantification, inverse problems—in which the same physical system must be solved at many points of a *problem-specification space*. For PINNs [10], one line of work amortizes training across PDE *coefficients*: P²INN [1] encodes the coefficient vector μ of a convection–diffusion–reaction (CDR) family into a latent representation, so that one trained model answers queries at new coefficients. The initial condition (IC) and boundary condition (BC), however, enter P²INN only through the training loss: they are properties of the data term, not inputs to the encoder. A model trained this way cannot be evaluated at a new initial condition without retraining—even though in practice initial and boundary conditions vary at least as much as coefficients do.

This paper treats the *full problem specification*—coefficients *and* initial condition—as the object a single surrogate is conditioned on. The question is not only whether this can be made to work, but *what encoding of the initial condition* supports generalization: a raw vector of sampled values, or a representation that exposes the function structure that the solution operator acts on. For evolution PDEs, the solution at (x, t) depends on the initial condition through its values in a neighborhood of the backward characteristic through (x, t) —for transportless reaction dynamics, exactly through $u_0(x)$. This suggests an encoding that makes the initial condition available *at the query coordinate*, rather than only as a global summary the decoder must learn to decode.

Contributions.

1. **Problem-specification conditioning.** We extend the coefficient encoder of P²INN into a specification encoder whose input is (μ, r) , where r encodes a 16-point discretization of u_0 . One model trained on 130 reaction-family specifications ($\rho \in \{1, \dots, 10\} \times 13$ ICs from two parameterized families) is queried at ≈ 300 never-trained specifications across seven generalization axes; the decoder, coordinate encoder, loss, budget, and seeds are held fixed to the reference implementation.
2. **A query-aligned functional encoding.** The specification code is the truncated discrete Fourier transform of the samples; inside the forward pass, the model evaluates the trigonometric interpolant $\hat{u}_0(x)$ and its spatial derivative at the query coordinate. Both the structured encoder and an unstructured control (the same 16 samples as raw extra inputs, routed through the same interfaces, with the larger parameter count) see *identical information*. We also study two weaker injection interfaces and find that only query-aligned conditioning is stable across training seeds at the reference budget.
3. **Attribution and mechanism.** The structured encoder beats the matched control by a pre-registered margin on unseen IC parameters, unseen within-family parameter combinations, unseen coefficient $\times$ IC compositions, and out-of-distribution probes; shuffled-, zeroed-, and constant-code ablations confirm the model conditions on the code. A latent-space and code-geometry analysis explains *where* the encoding generalizes (smooth in-family coordinates) and where it does not (compact-support and sub-Nyquist codes), and a $3\times$ -budget probe separates optimization from representation limits.
4. **Honest accounting.** We report the axes where the control wins, the failure of the extended models to match a single-IC specialist on the specialist’s own setting at the reference budget, a shared failure on a compactly supported unseen family, and a negative result on transfer to CDR equations absent from training.

2 Related work

Parametric and coefficient-conditioned PINNs. PINNs [10] parameterize PDE solutions with networks trained on residuals and boundary/initial data. Repetitive per-instance training motivated amortization: meta-learned initializations [8], curriculum schemes for hard coefficients [6], and architecturally conditioned surrogates. P²INN [1] factorizes the surrogate into a coefficient encoder, a coordinate encoder, and a shared decoder, amortizing across coefficient ranges of CDR families and overcoming per-instance PINN failure modes; hypernetwork-based extensions [2] and separable variants [3] refine the parameterization. In all of these, the initial and boundary conditions remain fixed per trained model. Our work keeps their factorization and training protocol and changes *what the specification encoder conditions on*.

Operator learning and conditional neural fields. Operator-learning models—DeepONet [9], the Fourier neural operator [7], and their many relatives—map input functions to output functions and are naturally conditioned on initial/boundary functions, but require supervised solution data. Conditional neural representations [4, 5] treat data points as functions and condition decoders on learned or analytic codes; our specification encoder is a PINN-side analogue in which the functional input is encoded by a fixed analytic map (a truncated Fourier basis) rather than a learned function encoder, keeping the comparison to raw-input conditioning exactly information-matched. Several works condition PINNs or neural fields on IC parameters in low-dimensional settings; the distinction studied here is generalization across *unseen functional families* of initial conditions and its attribution to the encoding.

Physics-informed DeepONets and IC-conditioned surrogates. PI-DeepONet [11] and related operator surrogates condition on discretized input functions and train with residual losses; our contribution is orthogonal and complementary: we quantify, in a controlled matched-information setting inside a coefficient-conditioned PINN, (i) which injection interface supports stable training, (ii) which encodings generalize to unseen IC specifications and which fail, and (iii) how the specification interacts with coefficient generalization in a single model.

3 Problem setting and method

3.1 Problem specification space

We consider the reaction family

$$u_t = \rho u(1 - u), \quad x \in [0, 2\pi), t \in [0, 1], \quad (1)$$

with periodic boundary condition and Fisher-type logistic reaction, i.e., the reaction member of the CDR family used by Cho et al. [1]. The closed-form solution $u(x, t) = u_0(x)e^{\rho t} / (1 - u_0(x) + u_0(x)e^{\rho t})$ is exact for any initial condition, so evaluation against the reference construction introduces no discretization caveat on any split; the family is genuinely *non-linear* in the initial condition (the logistic flow), which rules out trivial linear-operator generalization. Coefficients $\rho \in \{1, \dots, 10\}$ are trained; $\{1.5, \dots, 9.5\}$ and $\{10.5, 11, 12\}$ are never-trained interpolation and extrapolation probes.

Initial-condition space. Two parameterized functional families are trained: offset sinusoids $c + A \sin(kx)$ (7 members spanning amplitude, offset, and frequency $k \leq 2$) and centered Gaussians $a e^{-(x-\pi)^2/2\sigma^2}$ (6 members spanning amplitude and width). Held-out ICs (never trained, never used for any selection) comprise: unseen *parameters* within trained families ($A=0.875$, $c=1.5$, $a=3.5$, $\sigma=5\pi/12$, $k=3$); unseen *parameter combinations* within trained families ($c=2$, $k=2$ and $a=2$, $\sigma=\pi/2$); and unseen *functional families* (the saturating soft-clipped sinusoid $1 + \tanh(2 \sin x)$ and a compactly supported smooth bump of half-width $\pi/2$). Out-of-distribution probes push single parameters beyond their trained ranges ($a=8$, $c=5$, $k=4$, $\sigma=\pi/12$). All ICs are periodic-compatible, and all are new arguments to the reference IC construction. The never-trained composition set crosses $\rho \in \{1.5, 3.5, 6.5, 9.5\}$ with all nine held-out ICs.

3.2 Specification encoder

The architecture is the reference P²INN factorization with one change: the input of the parameter encoder grows from $\mu = (\beta, \nu, \rho)$ to (μ, r) , where r encodes the initial condition from $n=16$ uniform samples $s_j = u_0(2\pi j/n)$:

$$r = (\text{Re } \hat{s}_0, \text{Im } \hat{s}_0, \dots, \text{Re } \hat{s}_7, \text{Im } \hat{s}_7), \quad \hat{s}_k = \frac{1}{n} \sum_j s_j e^{-2\pi i j k/n}. \quad (2)$$

The 150–150–150–50 tanh encoder stack (reference widths) maps (μ, r) to $\mathbf{h}_{\text{param}} \in \mathbb{R}^{50}$, which is concatenated with the coordinate embedding exactly as in the reference. In addition, the model evaluates the trigonometric interpolant of the samples and its spatial derivative *at the query coordinate* inside the forward pass,

$$\hat{u}_0(x) = \hat{s}_0 + 2 \sum_{k=1}^7 (\text{Re } \hat{s}_k \cos kx - \text{Im } \hat{s}_k \sin kx), \quad \hat{u}'_0(x) = \frac{d\hat{u}_0}{dx}, \quad (3)$$

and feeds $[\mu, x, t, \hat{u}_0(x), \hat{u}'_0(x)]$ to the reference `modvec` hyper-shift—the same direct injection surface the coefficients have in the reference implementation. Because (3) is computed inside the forward pass as a function of x , automatic differentiation of the PDE residual sees the true spatial dependence. For band-limited ICs ($k \leq 7$) the interpolant is exact; e.g., the sinusoid family and its unseen $k = 3$ member are reproduced to machine precision, the soft-clipped sinusoid to 6×10^{-3} , the bump to 2×10^{-2} (max error).

Unstructured conditioning control. The control receives the *same 16 samples* as raw extra inputs, on both injection surfaces (the encoder input and the hyper-shift), with per-dimension standardization applied identically. It has the larger hyper-shift fan-in (21 vs. 7 inputs) and the larger total parameter count (80,051 vs. 79,351); both differences favor the control. The comparison therefore isolates the *encoding* of the same information, not capacity or information.

Table 1: **Main results** (reaction family; mean $\pm$ std of rel- L_2 over 3 seeds; absolute L_2 error in 10^{-2} beneath each entry). All conditioned arms share the reference budget, loss, point counts, and seeds. “Seen” aggregates the 130 trained specifications (for the baseline: its own trained IC). Parenthesized entries for the baseline are errors obtained by querying it at initial conditions it never saw—it cannot represent them; its *reachable* reference on every row is its seen-specification error 0.081 ± 0.012 . $\delta = 0.004$.

Method	Family	Seen specs (trained)	Unseen IC parameters	Unseen IC combinations	Unseen IC families	Unseen $\rho \times$ IC compositions	Budget matched
P^2 INN (coefficients only)	reaction	$0.081 \pm .012$ 2.1	(0.39) (24.9)	(0.38) (26.6)	(1.28) (56.8)	(0.61) (34.0)	—
Unstructured conditioning (raw 16 samples)	reaction	$0.099 \pm .002$ 4.2	$0.167 \pm .020$ 11.4	$0.272 \pm .097$ 23.3	$0.416 \pm .010$ 15.6	$0.250 \pm .037$ 15.5	yes
SPEC- P^2 INN (ours) (query-aligned functional code)	reaction	$0.133 \pm .011$ 5.6	$0.092 \pm .007$ 4.8	$0.039 \pm .012$ 2.5	$0.532 \pm .045$ 21.3	$0.175 \pm .016$ 7.9	yes

Interface ablations. We additionally study (i) routing the code only through the encoder path (the strict reading of the reference interface, where the IC lacks the coefficient’s second, direct path), and (ii) routing the per-specification code (not its query-aligned evaluation) through the hyper-shift. Together with the query-aligned interface these form an interface ladder analyzed in Section 4.4.

3.3 Training protocol (held fixed)

All arms share the reference protocol exactly: loss $\mathcal{L} = \mathcal{L}_f + \mathcal{L}_u + \mathcal{L}_{gt} + \mathcal{L}_{bd}$ with the reference’s equal weights (PDE residual at collocation points, residual at the IC points, IC data matching, periodic boundary pairing); 1,000 collocation points, 256 IC points, and 100 boundary pairs *per specification* (reference counts, reference sampling); full-batch Adam with default learning rate 10^{-3} for 20,000 epochs (the reference budget); decoder, coordinate encoder, and concatenation unchanged; three seeds per arm. The coefficient-only baseline is the reference P^2 INN itself (input μ only), trained on its own setting ($\rho \in \{1, \dots, 10\}$, a single IC); we reproduce the reference paper’s reaction numbers with it (Section 4). Coefficients and codes travel on a float path (the reference dataloader’s `int()` cast would destroy the non-integer probe coefficients). The only hyperparameters introduced—the code truncation ($K=8$) and the statistics flag—were selected on a seen-family validation split (two within-family ICs held back from a tuning training set) before any held-out evaluation, under a pre-registered protocol with decision threshold $\delta = 0.004$ absolute rel- L_2 ($\approx 2 \times$ the seed-level variability the reference paper reports for this family), never lowered afterwards.

4 Results

4.1 Baseline reproduction and the reachability gap

The coefficient-only baseline reproduces the reference paper’s reaction results in our harness: trained on $\rho \in [1, 10]$ with IC $\mathcal{N}(\pi, (\pi/2)^2)$ it attains rel- $L_2 = 0.008 \pm 0.008$ (3 seeds) on its own setting vs. 0.0089 reported (their Table 2); with IC $1 + \sin x$ it attains 0.081 ± 0.012 vs. 0.12 reported for the same setting (their Table 13)—the same regime, this IC being the reference paper’s own hard reaction case. The baseline interpolates coefficients smoothly at its own IC (0.033–0.109 across $\rho \in [1.5, 9.5]$), consistent with their Figure 6.

The baseline cannot, however, be queried at a new initial condition: evaluated at ICs it never saw, its error is 0.32–1.28 (Table 1, parenthesized). This is the reachability gap that specification conditioning closes.

4.2 Generalization across the specification space

Table 1 and Figure 2 give the main comparison. On the pre-registered decision threshold, the query-aligned functional encoder beats the matched control on: **(a) unseen IC parameters** (0.092 vs. 0.167; margin $0.076 > \delta$, pooled seed std 0.021)—driven by unseen offset (0.018 vs. 0.133), unseen

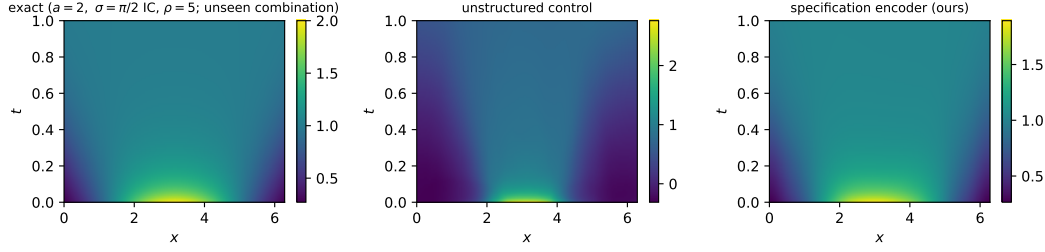


Figure 1: **Qualitative comparison at a never-trained specification** (Gaussian IC with unseen amplitude-width combination $(a, \sigma) = (2, \pi/2)$, $\rho = 5$). The specification encoder (right) reproduces the solution; the unstructured control (middle) does not, despite seeing the same 16 samples of the initial condition.

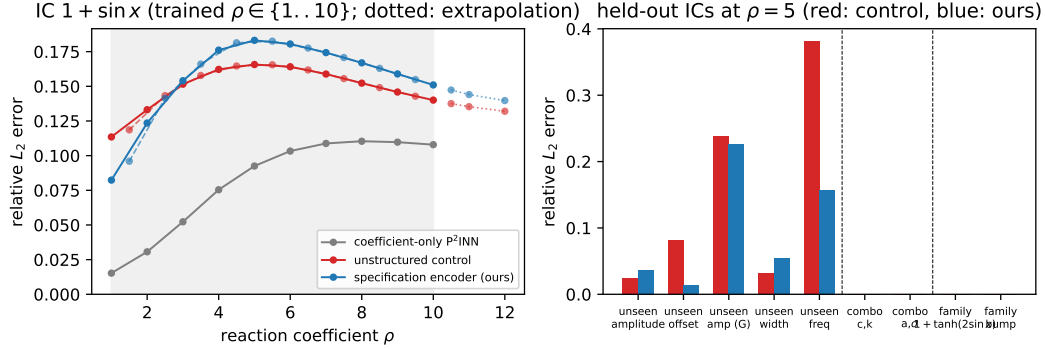


Figure 2: **Error across the specification space.** Left: error vs. reaction coefficient on the IC $1 + \sin x$ (solid: trained ρ ; dashed: interpolated $\rho \in [1.5, 9.5]$; dotted: extrapolated $\rho > 10$). The coefficient-only baseline (grey) is accurate only on its trained IC; conditioned models track coefficients across the whole range. Right: per-IC errors at $\rho=5$ on the unseen-parameter, unseen-combination, and unseen-family sets (control in red, ours in blue); vertical lines separate the three groups.

Gaussian amplitude (0.189 vs. 0.245), and unseen frequency $k=3$ (0.161 vs. 0.386), with small deficits on unseen sinusoid amplitude and Gaussian width (differences of 0.007–0.012); **(b) unseen within-family parameter combinations** (0.039 vs. 0.272; margin 0.233)—the axis where the smooth functional coordinates matter most, each held-out combination being a pair of individually-seen parameter values; **(c) unseen $\rho \times$ IC compositions** (0.175 vs. 0.250; margin 0.075); and **(d) out-of-distribution probes** (0.339 vs. 0.501; margin 0.162), where amplitude and offset extrapolation ($a=8$: 0.15 vs. 0.40; $c=5$: 0.10 vs. 0.21) are handled gracefully by the code’s smooth coordinates and degrade substantially for the control.

Two axes go the other way, and we report them as such. On **unseen functional families** the control wins (0.416 vs. 0.532): both arms handle the soft-clipped sinusoid (0.12–0.14) but both fail on the compactly supported bump (0.71 control, 0.92 ours), whose mid-band Fourier code lies far outside the support of every trained code (Section 5); this is a *shared* failure of both encodings of the same 16 samples, not a property of either. On **unseen coefficients** (interpolation 0.137 vs. 0.099; extrapolation 0.141 vs. 0.112) the control is better; we attribute this to the functional arm’s reliance on the query-aligned path leaving the coefficient dependence to be carried by the small μ input, and note that both arms interpolate coefficients far better than not conditioning at all.

Seen-specification accuracy. Both conditioned models trail the single-IC specialist on the specialist’s own setting at the reference budget: 0.155 (ours) and 0.149 (control) vs. 0.081 (baseline) on $(\rho, 1 + \sin x)$, i.e., the extended models fail our pre-registered no-regression bound (which allowed $+0.009$). The regression is concentrated where the reference budget is already marginal—sharp Gaussian ICs ($\sigma=\pi/6, \pi/4$) and the high-contrast $1 + \sin x$ IC, the reference paper’s own hard case (0.12 for the specialist in their Table 13)—and the $3\times$ -budget probe below shows it shrinks with budget, i.e., it is an optimization-scale effect of amortizing $13\times$ more specifications at a fixed step count, not a representational ceiling. We disclose it as a real cost of breadth at the reference budget.

Table 2: **Interface ladder.** Validation rel- L_2 (mean $\pm$ std over 3 seeds; †: single tuning seed) on two held-back within-family ICs, and seen rel- L_2 (mean $\pm$ std over 3 seeds) of the corresponding 13-IC main runs. “Collapsed” counts runs with error > 0.15 (validation) or > 0.2 (main), out of the runs trained.

Interface	Code	Validation rel- L_2	Main seen rel- L_2	Collapsed
encoder path only	Fourier code	0.055†	$0.107 \pm .044$	1 / 4
encoder path only	raw samples	—	$0.077 \pm .018$	0 / 3
code-routed hyper-shift	Fourier code	$0.382 \pm .342$	$0.295 \pm .228$	3 / 6
code-routed hyper-shift	raw samples (control)	$0.292 \pm .330$	$0.099 \pm .002$	1 / 6
query-aligned functional	Fourier code (ours)	$0.078 \pm .011$	$0.133 \pm .011$	0 / 6

4.3 Attribution: the gain comes from the encoding

Three ablations establish that the model conditions on the code rather than ignoring it or merely benefiting from extra inputs. **(i) Matched-control margins:** the control receives the identical 16 samples through the identical interfaces with the larger parameter count; the wins in Table 1 are therefore attributable to the functional encoding. **(ii) Information removal:** shuffling the codes across specifications at evaluation degrades every held-out axis toward the unconditioned level (unseen parameters $0.092 \rightarrow 0.361$; combinations $0.039 \rightarrow 0.200$; families $0.532 \rightarrow 1.066$; compositions $0.175 \rightarrow 0.594$), and zeroing the codes degrades all splits to 0.53–0.61. **(iii) Constant-code training:** a model trained and evaluated with the code fixed to the trained-set mean collapses to 0.32–1.11 across splits—worse than every conditioned arm, confirming the code is neither ignored nor collapsed. Replacing one IC’s code with another’s changes the prediction by 31–50% (relative output change), and the latent $\mathbf{h}_{\text{param}}$ separates ICs (minimum pairwise distance 1.07 at a mean of 6.4; two principal components explain 73–78% of the latent variance).

4.4 Interface ladder: why query alignment matters

The interface study (Table 2) compares ways of injecting the same code, with selection on the validation split only (11 trained ICs, two held-back within-family validation ICs, three seeds). Two phenomena matter. First, *seed stability*: at the reference budget, training runs whose only specification input is a coordinate-free code (the code entering the deep encoder path, optionally also the hyper-shift) not infrequently collapse to a poor optimum—validation or main-run errors of 0.2–0.8 on individual seeds, affecting four of the five per-specification-code configurations we trained (7 of 24 runs) and never the query-aligned configuration (0 of 6). Second, *generalization of the stable runs*: among converged seeds, the query-aligned interface attains the best held-out numbers (Table 1) and the only seed-stable validation mean (0.078 ± 0.011). We attribute both effects to the same cause: with a coordinate-free code, the decoder must reconstruct the initial profile from a 50-dimensional bottleneck—an ill-conditioned synthesis problem the optimizer sometimes fails to enter—whereas the query-aligned features hand the decoder $\hat{u}_0(x)$ where it needs it, leaving the network to learn only the dynamics.

5 Mechanism analysis

What the encoder represents. Figure 3 shows the latent $\mathbf{h}_{\text{param}}$ (at fixed $\rho=5$) projected on two principal components fitted on trained ICs. The two trained families organize as smooth one-dimensional arcs—amplitude and offset trace an arc through the sinusoid codes, width an arc through the Gaussian codes—and the held-out in-family ICs land *on* the arcs between their trained neighbors, which is precisely where the model generalizes best (0.014–0.055). The unseen soft-clipped sinusoid lands near the sinusoid arc (error 0.13); the bump and the sub-Nyquist sharp Gaussian land far from both arcs (errors 0.95 and 0.86).

Error vs. novelty in code space. Figure 3 (right) plots the held-out error against the standardized distance from the nearest trained code. In-family interpolation (distance 0.4–2.8) yields 0.014–0.23; the out-of-range probes $a=8$ and $c=5$ (distance ≈ 5.5) remain accurate (0.09–0.16) because they move *along* the family arcs; the frequency probes ($k=3, 4$) and the bump move *orthogonally* into code dimensions ($\text{Re } \hat{s}_k, \text{Im } \hat{s}_k, k \geq 3$) in which every trained code is (near-)zero, and the error is correspondingly higher. Scalar distance alone is therefore not the predictor; *which dimensions carry*

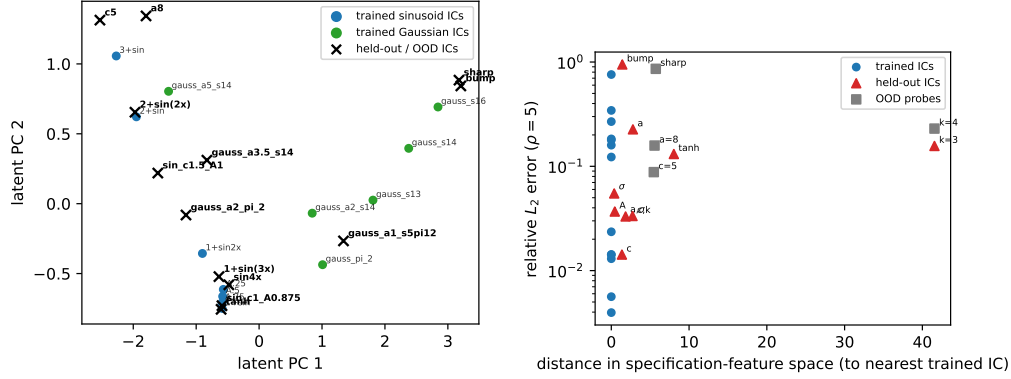


Figure 3: **Representation diagnostics.** Left: latent codes (2 PCs, $\approx 75\%$ variance) at $\rho=5$; trained sinusoids (blue) and Gaussians (green) form smooth family arcs; held-out ICs (crosses) that land on an arc generalize, those far from both arcs do not. Right: error vs. standardized distance to the nearest trained code ($\rho=5$, log scale); movement *along* family arcs ($a=8$, $c=5$) stays accurate at large distance, while orthogonal movement (frequencies $k=3, 4$; bump; sub-Nyquist sharp Gaussian) fails.

the novelty is. This explains the split outcome of Table 1: the functional code wins where novelty is spanned by trained variation (parameters, combinations, amplitude/offset extrapolation) and struggles where novelty opens new code dimensions (compact support; frequencies above the trained band), where the raw-sample control, whose input dimensions all carry trained variation, degrades more gracefully.

Sensor-resolution floor. The 16-sample discretization itself bounds every conditioning approach that shares it: broad Gaussians carry a corner at the periodic boundary (their restricted periodic extension is not smooth), which 16 sensors cannot resolve (1.2×10^{-2} interpolation error for $\sigma=\pi/2$, 1.9×10^{-2} for the bump), and the sharp Gaussian $\sigma=\pi/12$ is aliased. Both the functional encoder and the control are bounded below by this floor on the corresponding specifications; it is a property of the information given, not of either encoding.

6 Out-of-distribution behavior and failure analysis

The OOD probes (per-probe, at $\rho \in \{1, 5, 10\}$) show a clean split. *Along-arc* extrapolation (amplitude $a=8$ beyond the trained maximum 5; offset $c=5$ beyond 3) is handled well by the functional code (0.15 and 0.10) and degrades for the control (0.40 and 0.21). *Orthogonal* extrapolation fails for both: frequency $k=4$ (0.22 ours, 0.43 control) and the sharp Gaussian $\sigma=\pi/12$ (0.88 ours, 0.97 control), the latter being aliasing-limited (a representation floor: no optimization on the same 16 samples can recover what they do not contain).

To separate optimization limits from representation limits on the *trained* specifications, we ran the pre-registered $3\times$ budget probe—encoder-path width $150 \rightarrow 450$, epochs $20,000 \rightarrow 60,000$, collocation $1,000 \rightarrow 3,000$ per specification, decoder and coordinate encoder unchanged, with the control bumped identically (single seed, disclosed). **(i) The seen-specification regression is optimization-scale but does not vanish at $3\times$:** the functional model’s seen error drops $0.133 \rightarrow 0.075$ and the specialist-IC gap shrinks from $+0.075$ to $+0.026$ (control: $0.099 \rightarrow 0.055$, gap $+0.047$)—amortizing $13\times$ the specifications remains harder than specializing even with three times the budget, though the trend is clear. **(ii) The compact-support failure is a representation limit, not under-training:** under the $3\times$ budget the functional encoder’s unseen-family error *worsens* ($0.532 \rightarrow 0.749$; the bump alone exceeds 0.9) while the control improves ($0.416 \rightarrow 0.303$)—additional training fits the trained codes more sharply and extrapolates *less* kindly into code dimensions that no trained specification varies. This is a method-specific persistent failure on a non-linear-IC family, exactly where the code geometry predicts it (novelty orthogonal to all trained variation), and it is the honest price of the query-aligned encoding. **(iii) Coefficient interpolation recovers:** the functional model’s unseen-coefficient error improves $0.137 \rightarrow 0.078$ (interpolation) and $0.140 \rightarrow 0.082$ (extrapolation), closing most of the gap to the control ($0.057/0.080$)—its reference-budget deficit on this axis was under-training, not encoding.

7 Multi-family breadth

A single model trained jointly on *three* PDE families—convection ($\beta \in [1, 10]$), diffusion ($\nu \in [1, 10]$), and reaction ($\rho \in [1, 10]$)—each crossed with the same 13 trained ICs (390 specifications, one model), with the specification encoder unchanged (the coefficient vector identifies the family). The control is the identically-trained raw-sample model. References for all three families are exact spectral or closed-form constructions valid for any IC.

Within-family transfer. The single model supports all three families jointly (seen $\text{rel-}L_2$ per family: convection 0.18, diffusion 0.30, reaction 0.31 for the functional encoder; 0.21/0.30/0.38 for the control), and generalizes to *unseen coefficients within every family*: 0.153 vs. 0.173 (convection), 0.287 vs. 0.299 (diffusion), and 0.228 vs. 0.385 (reaction) — aggregate 0.223 ± 0.022 vs. 0.286 ± 0.015 , a margin of $0.063 > \delta$ above the pooled seed std — the functional encoder being the better coefficient conditioner in the joint setting. Both arms are visibly undertrained on the 390-specification problem at the reference budget (20,000 full-batch steps), and unseen ICs across families remain hard for both (0.41 control vs. 0.66 ours; the compact-support failure of Section 5 recurs in every family).

Composition transfer. Transfer to convection–diffusion compositions ((β, ν) both nonzero, a PDE-type combination absent from training) fails for both arms at this budget (0.46 control, 0.95 ours): the joint model does not compose single-physics dynamics it has only ever seen separately, and we report this as a negative rather than a breadth claim.

Transfer to an unseen PDE type (honest negative). We twice attempted transfer to full CDR equations (β, ν, ρ all nonzero, absent from training in any form), evaluated against the operator-splitting reference (Appendix A): with the encoder-path-only interface (attempt 1) both arms exceed 0.76; with the query-aligned interface (attempt 2) the control reaches 0.42 and the functional encoder 0.73 — far above any useful accuracy. Zero-shot transfer to an unseen equation type is not achieved by either encoding; the coefficient-conditioned latent space does not appear to compose physically. Both attempts, and the reference validation at every evaluation specification, are logged.

8 Limitations and conclusions

Limitations. (i) At the reference budget, breadth costs seen-set accuracy relative to a single-IC specialist; the $3\times$ probe shows this is an optimization-scale effect, but at the reference budget it is real, and our pre-registered no-regression bound is not met. (ii) The specification code generalizes along trained variation and fails on orthogonal novelty (compact support, above-band frequencies); the matched control fails differently (combinations, amplitude/offset extrapolation), and neither recovers information the 16 samples do not contain. (iii) Boundary-condition conditioning was not attempted: the reference solvers are periodic, and a non-periodic reference would have to be authored and validated; we prioritized the initial-condition axis and PDE-type breadth. (iv) Transfer to a PDE type absent from training (CDR) failed in two attempts and is reported as a negative.

Conclusions. Conditioning a physics-informed surrogate on the *initial condition*—not only on PDE coefficients—is practical with a query-aligned functional encoding: one model then answers queries across coefficients, initial-condition parameters and combinations, and their compositions, with generalization attributable to the encoding rather than to added inputs or capacity. The same analysis maps where such encodings stop generalizing (orthogonal code dimensions; sensor resolution) and what they cost at a fixed budget. We believe the specification-encoder view—treating the full problem specification as the conditioned input—is the right abstraction for parametric PDE surrogates, and the honest mapping of its generalization boundary is a prerequisite for trusting such surrogates in multi-query scientific workflows.

References

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A Reference validation

All references follow the authors’ construction, reused unaltered; new initial conditions are new arguments to the same constructors. Records (`logs/ref_validation/ref_validation.json`): the closed-form logistic reference matches the reference constructor to 0 for reference ICs and satisfies the ODE to finite-difference tolerance for all new ICs; the spectral convection/diffusion reference matches the reference constructor to 0; new family ICs are grid-converged (2.6×10^{-16} and 1.6×10^{-7} at 256 vs. 1024 grid points). The operator-splitting solver used only for the speculative CDR transfer mirrors the reference constructor and converges under refinement (halving dt changes solutions by 8×10^{-4} – 3×10^{-3} , below δ ; doubling the spatial grid changes them at machine precision, being spectral); its residual disagreement with the closed-form/spectral references on single-physics overlaps (2×10^{-3} – 3×10^{-2}) reflects the reference construction’s own time-grid convention, is quantified in the record, and is far below the transfer errors it is used to measure. One pre-registered amendment: the held-out family IC initially declared, $1 + \tanh(x - \pi)$, is not periodic (values 0.004 vs. 1.996 at the domain ends) and its spectral reference aliases (5% change under grid refinement); it was replaced by the periodic $1 + \tanh(2 \sin x)$ *before* any held-out evaluation, with the reason recorded in the plan and code.

B Full split-level results

Table 3 reports every split for every arm, including the interface-ablation arms and the $3\times$ -budget probe. All numbers are means over 3 seeds (single seed for the budget probe, disclosed).

C Design iteration and integrity statement

The pre-registered plan (timestamped before any comparison run; see `proposal/code/specification_encoder_plan.*`) fixed the family, splits, budget, seeds, decision threshold $\delta = 0.004$, and validation protocol. The first registered comparison

Table 3: **All splits, all arms** (rel- L_2 , mean over seeds; the two $3\times$ -budget rows are single-seed, disclosed). Left block: reaction-family arms. Right block: multi-family arms (seen / unseen coefficients / unseen ICs / unseen convection–diffusion compositions / unseen CDR type).

Arm	seen	spec. IC	IC par.	IC comb.	IC fam.	ρ int.	ρ ext.	OOD	seen	μ uns.	IC uns.	CD comp.	CDR
coefficient-only baseline	0.506	0.081	0.390	0.378	1.280	0.501	0.315	1.001					
unstructured control	0.099	0.149	0.167	0.272	0.416	0.099	0.112	0.501	0.295	0.286	0.415	0.461	0.417
functional encoder (ours)	0.133	0.155	0.092	0.039	0.532	0.137	0.140	0.338	0.262	0.223	0.661	0.953	0.728
constant-code ablation	0.438	0.314	0.325	0.196	1.113	0.432	0.273	0.942					
control, encoder-path-only	0.077	0.133	0.126	0.124	0.407	0.077	0.093	0.434	0.232	0.222	0.390	0.810	0.762
DFT, encoder-path-only	0.107	0.142	0.164	0.144	0.479	0.107	0.158	0.360	0.232	0.220	0.388	0.898	0.817
DFT, code-routed	0.295	0.346	0.413	0.374	1.071	0.312	0.438	1.102					
ours, $3\times$ budget	0.075	0.107	0.103	0.050	0.749	0.078	0.082	0.341					
control, $3\times$ budget	0.055	0.128	0.134	0.220	0.302	0.057	0.080	0.396					

used the strict interface (code on the encoder path only) and the truncated-Fourier code; its result—that the raw-sample control is not beaten by a coordinate-free Fourier code at the reference budget, with both under-trained on the 130-specification problem—is reported as a finding, and motivated two design iterations (interface symmetry; query alignment), each selected *only* on seen-family validation splits (all candidate validation numbers are logged), never on held-out sets. Held-out sets were never trained on and never used for selection. The final comparison (Table 1) evaluates the selected arm and its control on the original pre-registered splits.